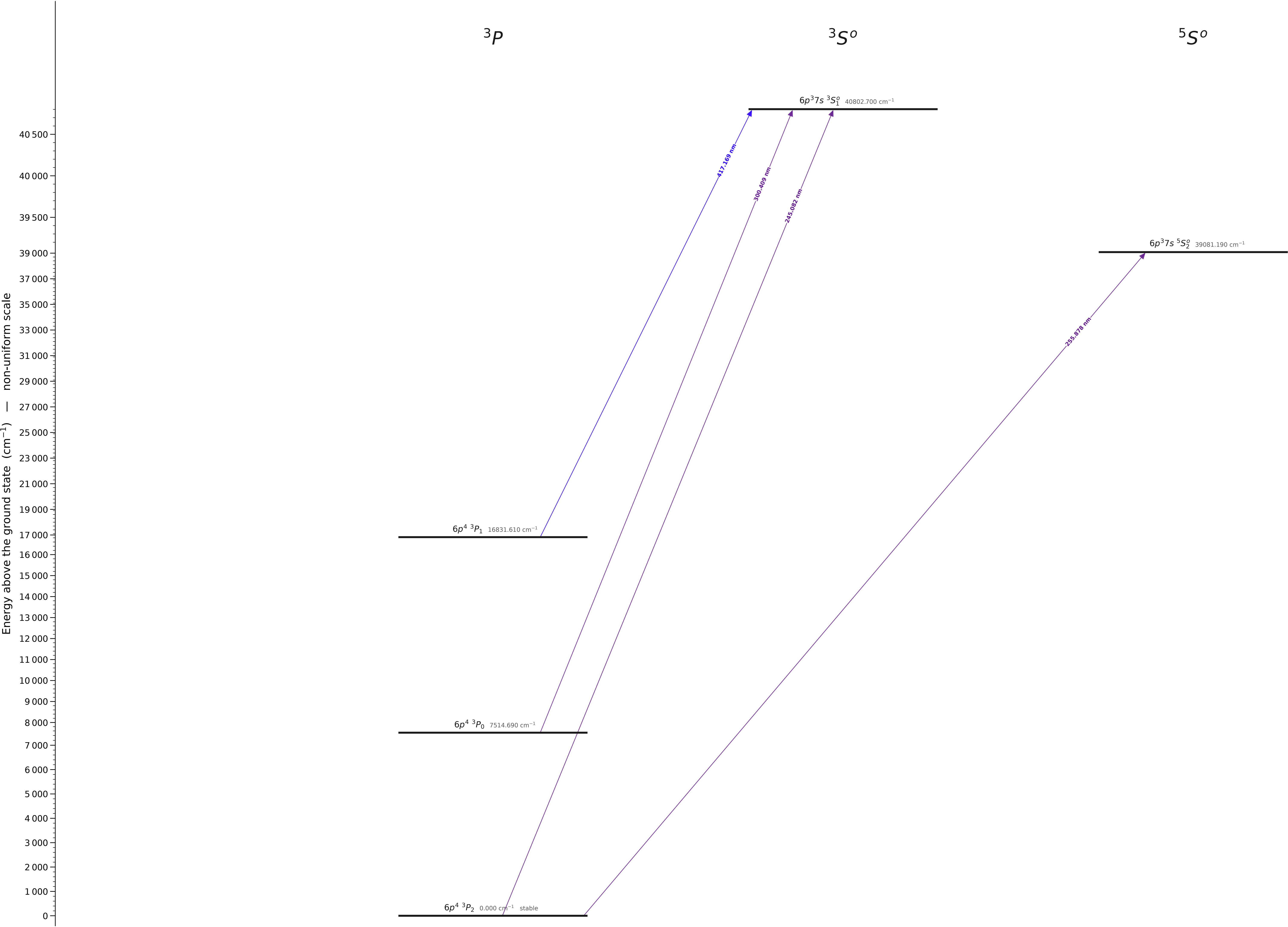
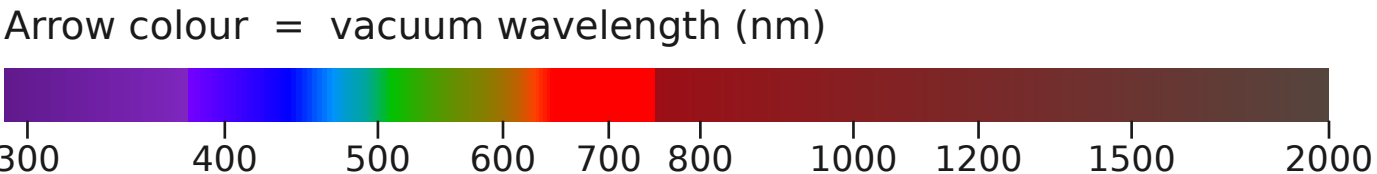


# Polonium

Po (Po I). Strongest classified transitions below 2 μm (NIST): 5 levels, 4 lines.

Po<sup>+</sup> ionisation limit 67896.31 cm<sup>-1</sup> = 8.41807 eV (147.283 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $| \langle J \| e r \| J' \rangle |$  in  $ea_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model or semi-empirical calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                       | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $  \langle J \  e r \  J' \rangle  $ ( $ea_0$ ) | A (s <sup>-1</sup> ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use                                                          |
|----------------------------------|--------------------|--------------------|-------------|-------------------------------------------------|----------------------|-------------------------|------------|--------------------------------------------------------------|
| $6p^4\ ^3P_2 - 6p^3 7s\ ^5S_2^o$ | 255.8776           | 255.8008           | 1171.6246   | -                                               | -                    | -                       | -          | first step of the ISOLDE-RILIS polonium resonance ionizat... |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

## <sup>209</sup>Po hyperfine structure ( $I = 1/2$ )

| Level              | A (MHz) | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|--------------------|---------|---------|--------------------------------------------------------|
| $6p^3 7s\ ^5S_2^o$ | 2127(3) | -       | F=3/2: -3190.50 F=5/2: +2127.00                        |

Measured A, B constants; sources in the data file.